

Heterogeneous Decentralized Fine-Tuning

PRIVACY-PRESERVING LLM ADAPTATION

THE PROBLEM

Data Governance

GDPR / HIPAA Compliance

Grok Chat Leak (2025)

370k+ private chats leaked via centralized server vulnerabilities.

HARDWARE MISMATCH

High-Tier

Rank 32

48GB VRAM

Edge-Tier

Rank 4

8GB VRAM

DECENTRALIZED
Federated Flow

THE SOLUTION

1

Reconstruct

$$\Delta W = B \times A$$

2

SVD Analysis

$$W = U \Sigma V^T$$

3

Project & Distill

Dynamic Rank Mapping